

Nate Posthumus

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PROFESSIONAL SUMMARY

Second-year mechanical engineering student at CU Boulder with hands-on experience designing and fabricating independent projects and writing control software. Seeking a summer engineering internship.

EDUCATION

University of Colorado Boulder
BS in Mechanical Engineering

May 2029

SKILLS

Fabrication & CAD: SolidWorks, 3D Printing, CNC
Programming: Python, C#, Linux Administration, OpenCV
Hardware: Electronics Prototyping, Motor Control
Languages: English, Spanish

ENGINEERING PROJECTS

Phone Puzzle-Solving Robot, Independent Project

Summer 2024 - Summer 2026

- Designed and fabricated a mechanical device that taps a phone screen to solve a puzzle game, actuated by stepper motors and a solenoid
- Implemented closed-loop encoder feedback for stepper motor positioning accuracy
- Implemented computer vision in Python with OpenCV to interpret and solve the puzzle from the phone screen in real time
- Wrote two serial-communicating scripts to coordinate puzzle solving and motor control
- Iterated through 3 prototypes independently, refining both mechanical design and software without instructor guidance

Raspberry Pi NAS, Independent Project

May 2026 - July 2026

- Built and deployed a self hosted raspberry pi server using OpenMediaVault and Docker in Linux
- Configured TailScale VPN for remote access to the server outside of the network

EXPERIENCE

Tennis Coaching Fellowship, Denver Tennis Park, Denver, CO

May 2025 - August 2025

- Coached students ages 8–16 in fundamentals, footwork, and match strategy
- Completed a total of 100 coaching hours, including 15 hours of unsupervised coaching

Volunteer Swim Coach, Special Olympics Swim, Denver, CO

May 2025 - August 2025

- Coached Special Olympics athletes across 2 seasons, assisting with swim meets
- Adapted coaching approach to meet the individual needs of each athlete

Lifeguard, Denver Parks and Recreation, Virginia Vale Swim Club

Summers, 2023 - Present

- Certified lifeguard for 3.5 years at two aquatic centers, a private club and a recreation center
- Monitor patron safety and make fast decisions in emergency situations

INTERESTS

Skiing, outdoor activities, martial arts, physical fitness